Extending the ADM formalism to Weyl geometry

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Outline

1. Introduction
2. Weyl Geometry
3. Decomposition in WIST
4. Outcome
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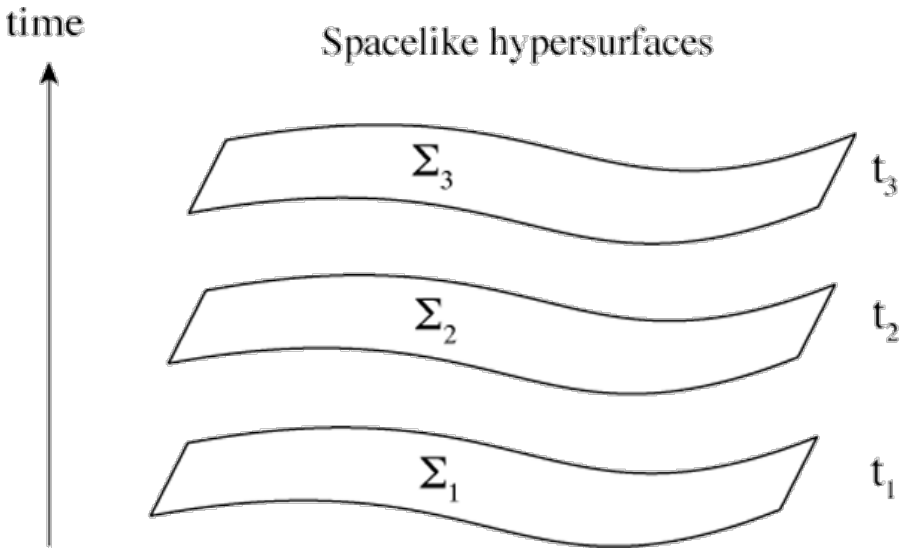
The road to quantum cosmology...

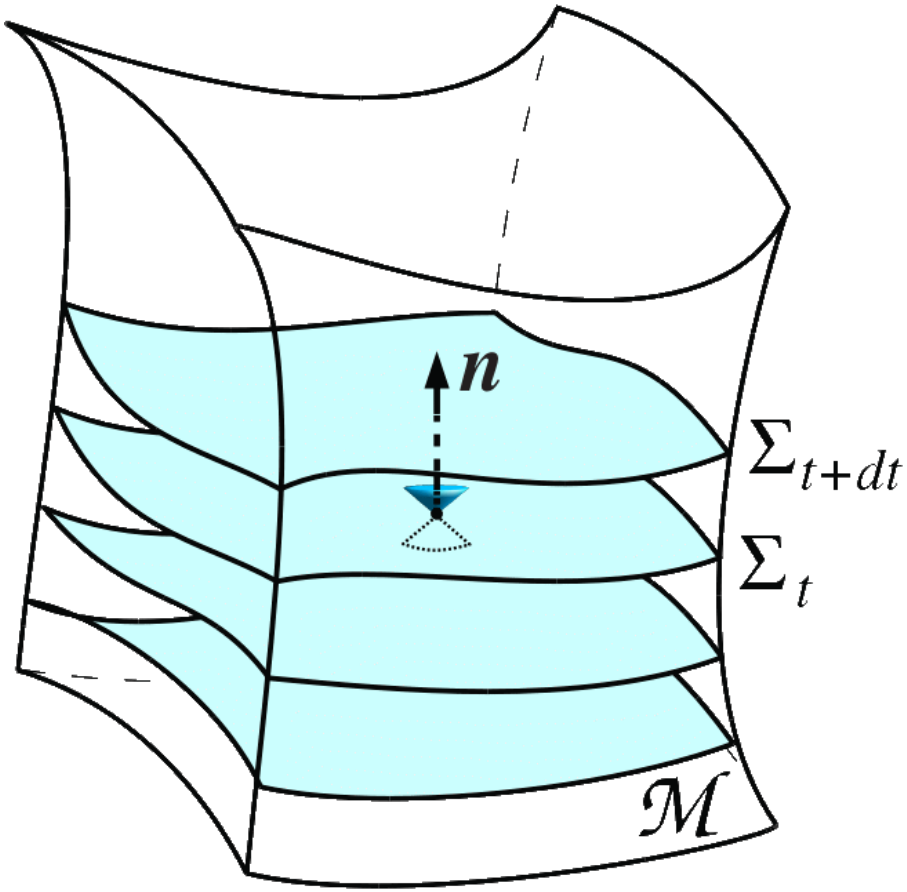
J.J. Halliwell, *Introductory Lectures on Quantum Cosmology* - Quantum Cosmology and Baby Universe

> *"Quantum cosmology is the application of quantum theory to the dynamical systems describing closed cosmology."*

Summary:

- Foliating (3+1)-spacetime in hypersurfaces $\Sigma(t)$;
- Gravity lagrangian in terms of intrinsic and extrinsic curvature;
- Legendre transformation to obtain the Hamiltonian





Splitting of the spacetime $\mathcal{M}$

A scalar field $t(x^\alpha)$ is defined in the manifold $\mathcal{M}$ such that

$$t = \text{constant}$$

describes a family of hypersurfaces space-like Σ_t.

This define a unit normal vector $n_\alpha \propto \partial_\alpha t$, which, to ensure proper normalization, it is introduced the *lapse function* N,

$$n_\alpha \doteq -N\partial_\alpha t \ \therefore \ n_0 = -N \ , \ n_i = 0$$

Spacetime decomposition $(3+1)$

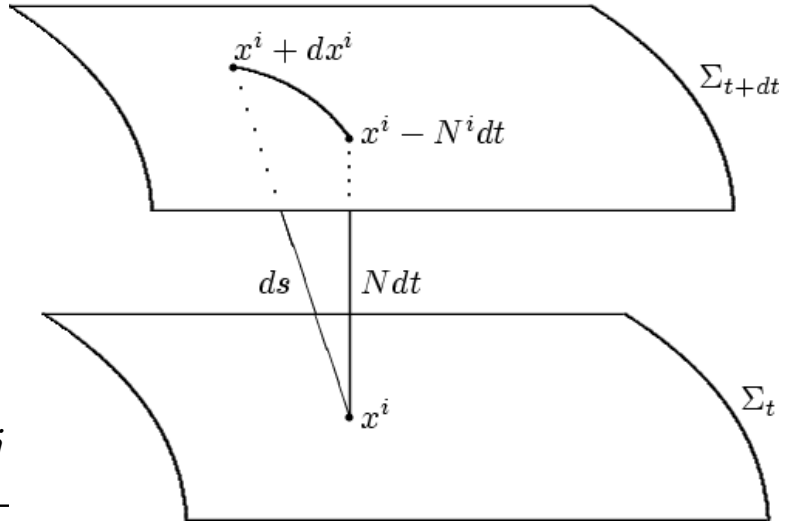
$$ds^2 = g_{\mu\nu}dx^\mu dx^\nu \equiv -\left(N^2 - N_i N^i\right)dt^2 + 2N_i dx^i dt + h_{ij}dx^i dx^j \qquad (1)$$

where we identify,

$$g_{00} = N^i N_i - N^2 \ , \ g_{0i} = N_i \ , \ g_{ij} = h_{ij}$$

with the contravariant components,

$$g^{00} = -N^{-2} \ , \ g^{0j} = \frac{N^j}{N^2} \ , \ g^{ij} = h^{ij} - \frac{N^i N^j}{N^2}$$



Furthermore, we have the projector tensor $h_{\mu\nu}$ defined as,

$$h_{\mu\nu} = g_{\mu\nu} + n_\mu n_\nu \implies h_{00} = N^i N_i \ , \ h_{0i} = N_i$$

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Weyl integrable manifold

C. Romero *et al*, *General relativity and Weyl geometry*, Class. Quantum Grav., 29, (2012)

This geometry is characterized by the nonmetricity condition

$$\nabla_\alpha g_{\mu\nu} = \phi_\alpha g_{\mu\nu}, \tag{2}$$

where ϕ is a scalar field, which we shall call *Weyl field*, and we define $\phi_\alpha \doteq \partial_\alpha \phi$.

The Eq.(2) is invariant under the simultaneous transformations

$$\bar{g}_{\mu\nu} = e^f g_{\mu\nu} \quad , \quad \bar{\phi} = \phi + f \tag{3}$$

and determines the affine connection (in this case we shall call *Weyl Connection*),

$$\Gamma^\alpha_{\mu\nu} = \left\{{}^{\alpha}_{\mu\nu}\right\} - \frac{1}{2} g^{\alpha\beta} \left(g_{\beta\mu}\phi_\nu + g_{\beta\nu}\phi_\mu - g_{\mu\nu}\phi_\beta \right). \tag{4}$$

Implications

Weyl frame: $(\mathcal{M}, g, \phi)$

Given a normal vector, for example n_μ, we will have

$$g_{\mu\nu} n^\mu n^\nu = -1 \quad \therefore \quad \nabla_\alpha \left(g_{\mu\nu} n^\mu n^\nu \right) = 0 \quad \Rightarrow \quad \boxed{n_\mu \nabla_\alpha n^\mu = \frac{1}{2} \phi_\alpha} \tag{5}$$

Likewise,

$$g^{\mu\nu} n_\mu n_\nu = -1 \quad \therefore \quad \nabla_\alpha \left(g^{\mu\nu} n_\mu n_\nu \right) = 0 \quad \Rightarrow \quad \boxed{n^\mu \nabla_\alpha n_\mu = -\frac{1}{2} \phi_\alpha} \tag{6}$$

since,

$$\nabla_\alpha g^{\mu\nu} = -\phi_\alpha g^{\mu\nu}. \tag{7}$$

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Covariant derivatives on Σ

Definition

We introduce a natural covariant derivatives that acts on 3-vectors, tangents to $\Sigma(t)$, by the projection of the $4D$ covariant derivative $\nabla_\mu X_\nu$, of any X_ν that satisfies $X^\alpha n_\alpha = 0$, in $\Sigma(t)$ using the projection tensor $h^\alpha_{\ \beta} = \delta^\alpha_{\ \beta} + n^\alpha n_\beta$:

$$D_\alpha X_\beta \doteq h^\mu_{\ \alpha} h^\nu_{\ \beta} \nabla_\mu X_\nu \tag{8}$$

Hence, assuming that D acts in scalar function via the rule $D_\alpha f = h^\beta_{\ \alpha} \nabla_\alpha f$, we determine the action of D in contravariants vectors,

$$D_\alpha V^\beta = h^\mu_{\ \alpha} h^\beta_{\ \nu} \nabla_\mu V^\nu \tag{9}$$

We can generalize to tensors of higher order, for example

$$D_\alpha T_{\mu\nu} = h^\beta_{\ \alpha} h^\gamma_{\ \mu} h^\lambda_{\ \nu} \nabla_\beta T_{\gamma\lambda} \tag{10}$$

Spatial nonmetricity

Using the above definition for the projector $h_{\mu\nu}$,

$$D_\alpha h_{\mu\nu} = h^\beta_{\ \alpha} h^\gamma_{\ \mu} h^\lambda_{\ \nu} \nabla_\beta h_{\gamma\lambda} \tag{11}$$

we get, since $h^\lambda_{\ \nu} n_\lambda = 0$ and $\nabla_\beta g_{\gamma\lambda} = \phi_\beta g_{\gamma\lambda}$,

$$D_\alpha h_{\mu\nu} = \varphi_\alpha h_{\mu\nu} \tag{12}$$

where $\varphi_\alpha \doteq h^\beta_{\ \alpha} \phi_\beta$ is the gradient Weyl field projected in the Σ.

Thus, if $\phi_\alpha = \phi_\alpha(t) \propto n_\alpha$, we will have that

$$D_\alpha h_{\mu\nu} = 0, \tag{13}$$

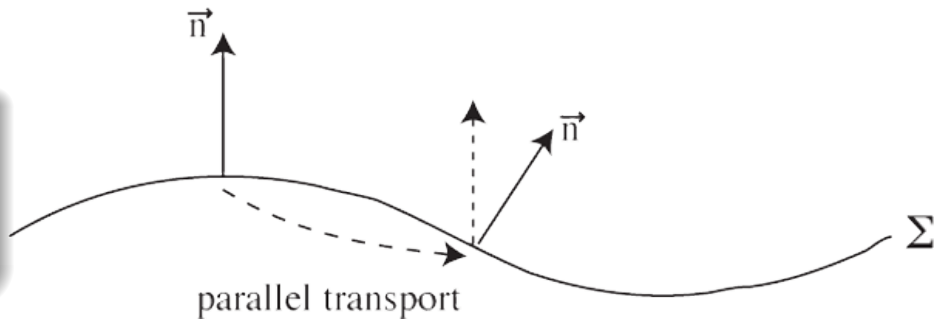
and, in this case, **the metricity applies on** $\Sigma(t)$.

Extrinsic curvature $\mathcal{K}_{\alpha\beta}$

Once that we have introduced the metric $h_{\mu\nu}$ and the spacial projection of the covariant derivative ∇_α on $\Sigma(t)$, we are able to describe the local properties of the spatial slices. So, to quantify how $\Sigma(t)$ it varies from event to event, we define the extrinsic curvature of $\Sigma(t)$ by

Definition

$$\mathcal{K}_{\alpha\beta} \doteq -h^{\mu}_{\ \alpha} h^{\nu}_{\ \beta} \nabla_\mu n_\nu \qquad (14)$$



As a result of nonmetricity, we can write

$$\mathcal{K}_{\alpha\beta} = -h^{\mu}_{\ \alpha}\left(\nabla_\mu n_\beta - \frac{1}{2}\phi_\mu n_\beta\right) \equiv K_{\alpha\beta} + \frac{1}{2}h_{\alpha\beta}\xi \qquad (15)$$

where $\xi \doteq n_\mu \phi^\mu$ is the normal component of the gradient Weyl field and $K_{\alpha\beta} \doteq -h^{\mu}_{\ \alpha} n_{\beta;\mu}$ is the riemannian extrinsic curvature.

Some properties of $\mathcal{K}_{\alpha\beta}$:

- $\mathcal{K}_{\alpha\beta}$ is spatial:

$$\mathcal{K}_{\alpha\beta}n^{\alpha} = \mathcal{K}_{\alpha\beta}n^{\beta} = 0 \quad , \quad \mathcal{K}^{\alpha\beta} \equiv h^{\alpha\mu}h^{\beta\nu}\mathcal{K}_{\mu\nu} \tag{16}$$

- $\mathcal{K}_{\alpha\beta}$ is symmetric:

$$\mathcal{K}_{\alpha\beta} = \mathcal{K}_{\beta\alpha} \tag{17}$$

- Defining a acceleration corresponding to the vector n_{μ} as $a_{\mu} \doteq n^{\alpha}\nabla_{\alpha}n_{\mu}$, we have:

$$-\nabla_{\beta}n_{\alpha} = \mathcal{K}_{\alpha\beta} + n_{\beta}a_{\alpha} - \frac{1}{2}n_{\alpha}\phi_{\mu}h^{\mu}{}_{\beta} \tag{18}$$

- The spatial components of $\mathcal{K}_{\alpha\beta}$ are given by

$$\boxed{\mathcal{K}_{ij} = \frac{1}{2N}\left[(D_iN_j + D_jN_i - h_{ij,0}) - (\phi_iN_j + \phi_jN_i - h_{ij}\phi_0)\right]} \tag{19}$$

Gauss-Codazzi equations

Given a particular foliation of the Weyl spacetime, $h_{\alpha\beta}$ and $\mathcal{K}_{\alpha\beta}$ contains the necessary information about the intrinsic and extrinsic properties of $\Sigma(t)$. So, in order to write the $4D$ curvature tensor in terms of $\mathcal{K}_{\alpha\beta}$ and the curvature of the 3-space, we start from the definition ${}^{(3)}R^{\alpha}_{\ \beta\mu\nu}$

Definition

$$D_\mu D_\nu X_\beta - D_\nu D_\mu X_\beta = {}^{(3)}R^{\alpha}_{\ \beta\mu\nu} X_\alpha \tag{20}$$

and we get the relation between $4D$ and $3D$ curvature tensors:

Gauss-Codazzi equations

$${}^{(3)}R_{\alpha\beta\mu\nu} = h^{\delta}_{\ \alpha} h^{\rho}_{\ \beta} h^{\sigma}_{\ \mu} h^{\gamma}_{\ \nu}\, {}^{(4)}R_{\delta\rho\sigma\gamma} + \mathcal{K}_{\alpha\mu}\mathcal{K}_{\beta\nu} - \mathcal{K}_{\alpha\nu}\mathcal{K}_{\beta\mu} \tag{21}$$

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Now that we have made the appropriate generalizations of geometric quantities for the Weyl integrable manifold, we can obtain the ADM lagrangian to any Weyl-type gravity theory.

For instance we have,

Weyl integrable spacetime **(WIST)**, by M. Novello *et al.*:

$$\mathcal{S}_{wist} = \int \sqrt{-g} d^4x \left(R + \xi \phi_\alpha \phi^\alpha\right) \tag{22}$$

and

geometric Brans-Dicke **(gBD)**, by T. Almeida *et al.*:

$$\mathcal{S}_{gBD} = \int \sqrt{-g} d^4x e^{-\phi} \left(R + \omega \phi_\alpha \phi^\alpha\right) \tag{23}$$

where the R is the curvature scalar of Weyl geometry, ϕ is the Weyl field itself and $e^{-\phi} = \Phi_{BD}$.

R in terms of the extrinsic and intrinsic curvatures

By contraction of $R_{\lambda\nu\sigma\tau}$ with $h_{\alpha\beta}$ in order to obtain the curvature scalar R, we get

$$R = h^{\lambda\sigma} h^{\nu\tau} R_{\lambda\nu\sigma\tau} - 2n^{\lambda} n^{\sigma} R_{\lambda\sigma} \tag{24}$$

at the end of the day, we have that

$$\boxed{R = {}^{(3)}R - \mathcal{K}^{\gamma}_{\ \nu}\mathcal{K}^{\nu}_{\ \gamma} + \mathcal{K}^2 + 2\left(\nabla_{\mu} + \phi_{\mu}\right) A^{\mu}} \tag{25}$$

where $A^{\mu} \doteq a^{\mu} + n^{\mu}(\mathcal{K} - \frac{1}{2}\xi)$.

Divergence term:

Note that,

$$\nabla_\alpha A^\alpha = A^\alpha_{;\alpha} - \frac{1}{2}\left(\delta^\alpha_\alpha \phi_\beta + \delta^\alpha_\beta \phi_\alpha - g_{\alpha\beta}\phi^\alpha\right) A^\beta. \tag{26}$$

Thus,

$$A^\alpha_{;\alpha} = \left(\nabla_\alpha + 2\phi_\alpha\right) A^\alpha. \tag{27}$$

This way, we can write

$$R = {}^{(3)}R - \mathcal{K}^\gamma_\nu \mathcal{K}^\nu_\gamma + \mathcal{K}^2 - 2\phi_\mu A^\mu + 2\left(\nabla_\mu + 2\phi_\mu\right) A^\mu \tag{28}$$

where the latter term is a *Surface Term.*

Therefore, we define

$$\boxed{{}^{w}L_{ADM} = {}^{(3)}R - \mathcal{K}^\gamma_\nu \mathcal{K}^\nu_\gamma + \mathcal{K}^2 - 2\phi_\mu A^\mu} \tag{29}$$

Thanks!

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